mmIR: Frequency-Space Inverse Rendering for 3D Millimeter-Wave Radar ADC Synthesis

Anonymous ECCV 2026 Submission

Paper ID #11932

Abstract. High-resolution 3D radar data is scarce: commodity mmWave sensors use small antenna arrays that limit angular resolution to several degrees, and existing datasets provide only 2D range–azimuth maps or sparse point clouds rather than raw ADC. Hardware scaling is expensive, synthetic-aperture scanning is impractical at fleet scale, and learned synthesis methods are bottlenecked by the very data shortage they aim to address. We present mmIR, an open-source differentiable FMCW radar inverse renderer that fits a physics-based forward model to real captures and re-renders from dense virtual apertures to synthesize high-resolution 3D radar data. Using LiDAR-derived meshes as a geometric scaffold, mmIR optimizes per-vertex ITU physics materials, vertex normals, and antenna beam patterns through end-to-end automatic differentiation of a phase-coherent MIMO forward model with multi-bounce propagation, polarization, and free-space diffraction. On seven outdoor ColoRadar scenes, mmIR achieves 0.63 mean Pearson correlation on range–azimuth maps versus 0.32 for Sionna-RT. Scenes trained on a cascaded imaging radar transfer to a co-located single-chip radar without re-training (0.58 correlation), and dense virtual arrays (100×100 elements) produce single-frame 3D occupancy validated against LiDAR.

Keywords: Differentiable rendering · FMCW radar · Inverse rendering · mmWave · 3D reconstruction

1 Introduction

Millimeter-wave (mmWave) radar penetrates fog, rain, and airborne dust [16], measures Doppler velocity, and returns strong echoes from metals, yet it remains underutilized for 3D scene understanding. Commercial single-chip radars have only tens of antennas, limiting angular resolution to several degrees [14], and their axis-biased layouts yield 2D range–azimuth (RA) slices rather than full 3D range–azimuth–elevation (RAE) volumes. Public datasets further limit access by providing only post-processed outputs (RA maps or sparse point clouds) rather than raw ADC [4, 9]. The result is a shortage of high-resolution, 3D, raw FMCW ADC data: the fundamental bottleneck for learning-based radar perception. Detection, segmentation, occupancy prediction, and super-resolution all require training signal that current sensors and datasets cannot supply, and addressing this gap would enable these applications.

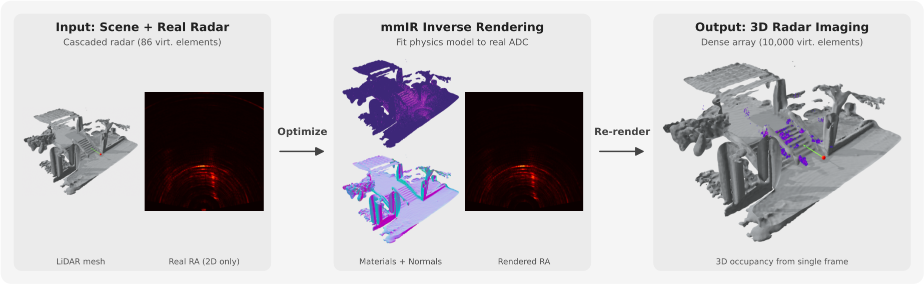

Fig. 1: Left: LiDAR mesh and real 2D range–azimuth (RA) map from a commodity cascaded radar. **Center:** mmIR optimizes per-vertex materials, normals, and antenna beam patterns by fitting rendered ADC to real measurements. **Right:** The optimized scene is re-rendered from a dense 100×100 virtual array for single-frame 3D occupancy detection.

Existing approaches each fall short. Hardware scaling [42] increases cost without addressing data diversity. SAR [31, 49, 56] requires mechanical scanning impractical at fleet scale. Neural and Gaussian synthesis methods [8, 19, 27, 36, 43] are bottlenecked by the very data scarcity they aim to address: models trained on low-resolution 2D RA maps cannot recover 3D structure that was never observed.

Physics-based ray tracing (SBR) for mmWave is well established [13, 18], but existing pipelines are forward-only: geometry and materials must be specified *a priori*, with no mechanism to calibrate against real measurements. Differentiable rendering has closed this gap for RGB [29, 55], LiDAR [23], and acoustics [22], and emerging RF methods take initial steps. Sionna RT [18] exposes gradients for antenna patterns and scattering coefficients but not through ray tracing itself (no geometry or normal gradients) and does not model wideband FMCW signals. Hofmann et al. [17] estimate SFCW materials but assume known geometry with single-bounce propagation, and InverTwin [10] addresses multipath gradient pathologies but validates on synthetic data only. What is missing is an end-to-end differentiable FMCW renderer that can learn scene parameters from real radar captures without requiring ground-truth materials.

We present mmIR, an open-source differentiable FMCW radar inverse renderer that fits a physics-based forward model to real captures and re-renders from dense virtual apertures to synthesize high-resolution 3D radar data. Radar resolution is too coarse to reconstruct geometry directly, so we use LiDAR-derived meshes [24] as a geometric scaffold, a practical choice given the far greater availability and resolution of LiDAR data. Inverse rendering then recovers the electromagnetic properties: mmIR optimizes per-vertex ITU physics materials (permittivity, roughness, correlation length, blend factor, slab thickness), vertex normals, and antenna beam patterns through end-to-end automatic differentiation of a phase-coherent MIMO forward model with multi-bounce propagation, polarization-aware BSDFs, specular manifold sampling [52], and free-space diffraction [41]. The pipeline also supports differentiable geometry and 6-DoF pose gradients with projective boundary sampling [53], though we keep these fixed in our experiments because LiDAR-radar alignment preprocessing (Sec. B.2 in the supplement) already provides accurate geometry and pose. After fitting, the optimized scene is frozen and re-rendered from arbitrarily large virtual apertures to synthesize dense, physics-consistent raw ADC (Fig. 1), enabling downstream applications such as 3D occupancy detection, radar super-resolution, and synthetic training data generation that would otherwise require prohibitively large physical arrays.

On seven outdoor ColoRadar [26] scenes, mmIR achieves 0.63 mean Pearson RA correlation versus 0.32 for Sionna-RT. We demonstrate cross-sensor transfer: scenes trained on a cascaded imaging radar (12TX$\times$16RX, 86 virtual elements) produce valid ADC for a co-located single-chip radar (3TX$\times$4RX, 12 virtual elements) without re-training (0.58 correlation), providing evidence that the optimized scenes capture transferable physics rather than sensor-specific patterns. We further show single-frame 3D occupancy detection from dense virtual arrays (100TX$\times$100RX) validated against LiDAR. Our contributions are:

1. To the best of our knowledge, the first open-source, end-to-end differentiable inverse renderer for wideband FMCW radar, enabling joint optimization of per-vertex materials, normals, and antenna patterns directly from real radar captures, with support for multi-bounce propagation, specular manifold sampling, polarization-aware BSDFs, free-space diffraction, and differentiable geometry and pose gradients with projective boundary sampling.
2. A virtual-aperture re-rendering framework that synthesizes high-resolution 3D raw ADC from fitted scenes via arbitrarily large synthetic arrays, enabling 2D-to-3D lifting and 3D occupancy detection.
3. Cross-sensor transfer validation on real COTS radar, demonstrating that scenes optimized on one sensor generalize to a different configuration without re-training.

1.1 Scope

The primary goal of mmIR is high-quality ADC radar data synthesis: by fitting a differentiable forward model to real radar captures on real LiDAR-derived geometry, mmIR produces dense, physics-consistent raw ADC that would otherwise require expensive hardware arrays to acquire. We envision practitioners using optimized scenes to generate large-scale datasets for training downstream perception models (detection, segmentation, occupancy prediction) without additional data collection. We validate on real measurements from commercial off-the-shelf (COTS) radars, specifically the Texas Instruments MMWCAS-RF-EVM cascade and AWR1843 single-chip sensors, demonstrate cross-sensor transfer, and show that the synthesized data is sufficient for single-frame 3D occupancy detection from dense virtual arrays (Sec. 4.3). Our experimental evaluation compares against open-source, physics-based ray-tracing inverse rendering methods (Table 1), ensuring reproducibility and a like-for-like comparison of explicit forward models.

Table 1: Feature comparison of physics-based differentiable radar renderers. Signal: ADC = raw FMCW, CIR = channel impulse response, IF = intermediate-frequency phasors. Multi-b. = multi-bounce; Diffr. = edge diffraction; Polar. = polarization; Geom. = differentiable geometry; Norm. = differentiable normals; Mat. = physics material model; Per-vtx = per-vertex materials; Bound. ∇ = boundary gradients; Real = validated on real data. Checkmarks denote supported pipeline capabilities; see Sec. B.4 in the supplement for which parameters are active in our experiments.

Method	Open	Repr.	Signal	Multi-b.	Diffr.	Polar.	Geom.	Norm.	Mat.	Pose	Beam	Per-vtx	Bound. ∇	Real
Sionna RT [18]	✓	Mesh	CIR	✓	✓	✓	–	–	No BRDF	✓	–	–	–	–
SFCW Inv. [17]	✓	Mesh	IF	–	–	✓	–	Shading	✓	Pos.	–	✓	–	✓
InverTwin [10]	–	Mesh	CIR	✓	✓	–	✓	–	No BRDF	✓	–	–	✓	✓
mmIR (Ours)	✓	Mesh	ADC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

2 Related Works

Differentiable rendering beyond optics. Differentiable rendering couples explicit scene parameterizations with gradient-based optimization to recover geometry, materials, and sensor parameters [15, 39, 46, 47]. Core advances include analytic inverse rendering and MC estimator reparameterizations [2, 29, 33], as well as practical path tracing systems and neural/hybrid representations [1, 3, 21, 35, 37, 55]. These methods operate almost exclusively in the optical regime. In RF, Sionna RT [18] propagates gradients through field coefficients and CIR computation but not through path topology: its path solver runs under gradient suspension and subsequently detaches all path geometry, so vertex positions and surface normals receive no gradients, restricting inverse rendering to quantities that enter only the CIR computation (antenna patterns, scattering coefficients, TX/RX positions). It also does not model wideband FMCW signals; Hofmann et al. [17] estimate SFCW materials but require known geometry, model only single-bounce propagation, and treat visibility as non-differentiable; and InverTwin [10] addresses multipath gradient pathologies via path-space differentiation but substitutes a Gaussian kernel surrogate for radar range processing and validates on synthetic data only. mmIR bridges this gap with end-to-end AD through ray tracing itself, enabling gradients to flow through all bounces back to geometry, materials, and sensor parameters directly from raw frequency-space FMCW ADC (Table 1).

Ray tracing for FMCW signal modeling. Radar simulation typically employs shooting-and-bouncing rays (SBR) to scale beyond full-wave solvers, producing path geometry and attenuation for large scenes and MIMO arrays [13]. Recent computational RF imaging further demonstrates multi-bounce physics for reconstruction on real mmWave hardware [12]. However, these pipelines are forward-only: scene and sensor parameters are fixed, precluding calibration against real captures. mmIR makes SBR-style FMCW rendering fully differentiable, exposing gradients w.r.t. mesh geometry, per-vertex materials, radar pose, and antenna patterns to close the sim-to-real gap.

Neural and Gaussian methods for radar data synthesis. Neural fields have been adapted to radar across modalities: SAR/ISAR shape reconstruc-

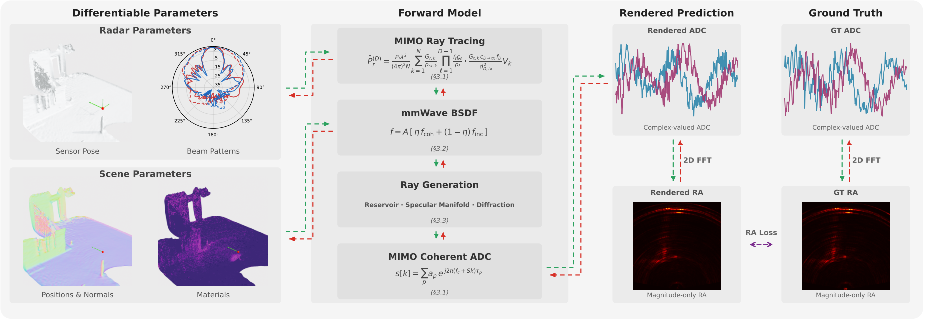

Fig. 2: Overview of the mmIR pipeline. Given a LiDAR-derived mesh and initial sensor parameters, the differentiable forward model traces phase-coherent MIMO paths via multiple importance sampling over reflection and diffraction for all TX–RX pairs, then synthesizes raw FMCW ADC. End-to-end automatic differentiation through the full path (ray tracing → BSDF → phasor accumulation → ADC) enables joint optimization of per-vertex materials, geometry, normals, and 6-DoF radar pose from a range–azimuth reconstruction loss.

tion [11, 28], FMCW occupancy and Doppler recovery [19, 43], frequency-space response synthesis [8, 54], and multi-modal fusion with RGB/LiDAR [34, 38]. RadarSplat [27] provides a Gaussian splatting [25] alternative for range–azimuth rendering. These methods learn opaque, sensor-specific representations: DART [19] discards phase entirely (magnitude-only range–Doppler supervision), none expose physically interpretable material parameters, and none model multi-bounce propagation, polarization, or diffraction. Moreover, methods that render post-processed images directly must learn to reproduce FFT point-spread artifacts (sidelobes, windowing); by rendering raw ADC, mmIR applies identical FFT processing to both rendered and measured data, cleanly decoupling scene physics from signal processing. mmIR maintains an explicit mesh with per-vertex ITU materials and polarization-aware BSDFs, enabling joint scene–sensor calibration and re-rendering from arbitrary virtual apertures for 3D imaging.

3 Methods

We model radar sensing as a forward rendering problem (Fig. 2): trace multi-bounce rays through a scene, evaluate surface interactions via a physically-based mmWave BSDF, and accumulate phase-coherent paths into raw FMCW ADC samples that can be directly compared against real captures. Sec. 3.1 derives the forward model from the bistatic radar equation through multi-bounce Monte Carlo estimation to ADC synthesis; Sec. 3.2 defines the mmWave BSDF governing surface interactions; Sec. 3.3 describes ray generation strategies; and Sec. 3.4 formulates the differentiable inverse rendering pipeline.

Our inputs are synchronized radar–LiDAR data from ColoRadar [26]. LiDAR frames are aggregated into a triangle mesh via screened Poisson reconstruction [24], and a GPU-accelerated coarse-to-fine alignment corrects residual sensor misregistration (Sec. B.1–B.2 in the supplement). FMCW signal and TDM-MIMO imaging derivations are in the supplement (Sec. A–A.3).

3.1 Forward Model

Our forward model is grounded in the bistatic radar equation. For a single TX–RX pair observing a point target at surface point $\mathbf{x}$ with normal $\mathbf{n}$, at distances d_t, d_r from transmitter and receiver, the received power P_r is:

$$P_r = \frac{P_t\, G_t\, G_r\, \lambda^2\, \sigma}{(4\pi)^3\, d_t^2\, d_r^2}, \tag{1}$$

where P_t is transmit power, G_t, G_r are directional antenna gains, λ is wavelength, and σ is the bistatic radar cross-section. Replacing the point-target RCS with a spatially varying BSDF $f(\boldsymbol{\omega}_t, \mathbf{x}, \boldsymbol{\omega}_r)$, where $\boldsymbol{\omega}_t$ and $\boldsymbol{\omega}_r$ are unit directions toward the transmitter and receiver, and converting from surface area element dA_x to receiver solid angle via $d\omega_r = (c_r/d_r^2)\, dA_x$ with $c_r = \max(0, \mathbf{n}\cdot\boldsymbol{\omega}_r)$ (derivation in the supplement, Sec. A.4), the total received power integrated over the receiver hemisphere Ω_{rx} becomes:

$$P_r = \frac{P_t\, \lambda^2}{(4\pi)^2} \int_{\Omega_{rx}} \frac{G_t(\boldsymbol{\omega}_t)\, G_r(\boldsymbol{\omega}_r)\, f(\boldsymbol{\omega}_t, \mathbf{x}, \boldsymbol{\omega}_r)\, c_t\, V}{d_t^2}\, d\omega_r, \tag{2}$$

where $c_t = \max(0, \mathbf{n}\cdot\boldsymbol{\omega}_t)$ is the cosine factor and $V \in \{0, 1\}$ is visibility. This solid-angle formulation directly motivates RX-centric sampling: trace rays from the receiver, intersect the scene, and evaluate the integrand with a deterministic connection to the transmitter.

For multi-bounce propagation with path depth D and interaction points $\mathbf{x}_1, \ldots, \mathbf{x}_D$, each stochastically sampled segment contributes a BSDF-over-PDF ratio and geometric coupling. Given N Monte Carlo samples, the unbiased estimator is (derivation in the supplement, Sec. A.4):

$$\hat{P}_r^{(D)} = \frac{P_t\, \lambda^2}{(4\pi)^2}\, \frac{1}{N} \sum_{k=1}^{N} \frac{G_{r,k}}{\rho_{rx,k}} \prod_{\ell=1}^{D-1} \frac{f_\ell\, \mathcal{G}_\ell}{\rho_\ell} \cdot \frac{G_{t,k}\, c_{D\to tx}\, f_D}{d_{D,tx}^2}\, V_k, \tag{3}$$

where $\mathcal{G}_\ell = c_{\ell\to\ell+1}\, c_{\ell+1\to\ell} \,/\, d_{\ell,\ell+1}^2$ is the geometric coupling between consecutive bounces, with $c_{\ell\to\ell+1} = \max(0, \mathbf{n}_\ell \cdot \boldsymbol{\omega}_{\ell\to\ell+1})$ the cosine factor at bounce ℓ toward bounce $\ell{+}1$ and $d_{\ell,\ell+1}$ the distance between them. The BSDF at each bounce is $f_\ell = f(\boldsymbol{\omega}_{\ell-1}, \mathbf{x}_\ell, \boldsymbol{\omega}_{\ell+1})$ (Sec. 3.2), which decomposes each surface interaction into slab transmission, coherent specular reflection, incoherent diffuse scattering, and edge diffraction. The final term connects deterministically to the transmitter: $c_{D\to tx}$ is the cosine factor at $\mathbf{x}_D$ toward the TX and $d_{D,tx}$ is their distance. The sampling PDFs ρ_ℓ are determined by the ray generation strategies in Sec. 3.3. The single-bounce case ($D{=}1$) reduces to $G_t\, G_r\, c_t\, f\, V \,/\, (d_t^2\, \rho_{rx})$, and we use Russian roulette [46] for unbiased path termination.

The MC estimator provides a complex amplitude for each traced path. To synthesize raw FMCW ADC, each path p for TX i and RX j contributes a phase-delayed chirp sample using the complex baseband form (Sec. A.1–A.2 in the supplement):

$$s_{ij}^{(p)}[k] = a_{ij}^{(p)} \exp\Big(j\, 2\pi (f_c + S t_k)\, \tau_{ij}^{(p)}\Big), \tag{4}$$

where k indexes fast-time ADC samples at times t_k, j is the imaginary unit, $\tau_{ij}^{(p)} = R_{ij}^{(p)}/c$ is the round-trip delay for total path length $R_{ij}^{(p)}$ at speed of light c, f_c is the carrier frequency, S is the chirp slope, and $a_{ij}^{(p)} \in \mathbb{C}$ aggregates the MC throughput from Eq. (3): antenna gains, BSDF evaluations, geometric coupling, and spreading losses. The rendered ADC sums coherently over all paths: $s_{ij}[k] = \sum_p s_{ij}^{(p)}[k]$.

Equations (1)–(4) describe a single TX–RX pair. In a MIMO array, each pair is evaluated independently through the same forward model; however, coherent MIMO imaging via FFT [30] requires exact relative phase across all pairs, so independent MC samples per receiver would corrupt this. Following the MIMO ray reuse strategy of Schüßler et al. [40], we share path topology across all TX–RX pairs: a single set of interaction points is found from shared seeds, then evaluated for all pairs by vectorizing only the TX$\to$ $\mathbf{x}_1$ and $\mathbf{x}_D$ $\to$RX legs. Each pair produces unique amplitudes and phases but shares bounce topology, preserving MIMO coherence while reducing cost from $\mathcal{O}(N_{\mathrm{TX}} \cdot N)$ to $\mathcal{O}(N)$.

3.2 Physically-Based mmWave BSDF

The BSDF f appearing in the MC estimator (Eq. (3)) governs how electromagnetic energy is redistributed at each surface interaction. We model it using ITU-R P.2040 recommendations for mmWave propagation [20], inspired by the coherent spatially varying BSDF (CSVBSDF) formulation of Wei et al. [50]. Each mesh vertex stores six physically interpretable material parameters: real and imaginary relative permittivity ($\varepsilon_r', \varepsilon_r''$), RMS surface roughness σ_h, correlation length l_c, a KA/SPM blend factor τ, and slab thickness d. At each ray–triangle intersection, parameters are barycentric-interpolated from the three triangle vertices ($\mathbf{m}_i \in \mathbb{R}^6$), ensuring C^0-continuous material fields with gradients flowing to all three vertices.

The BSDF decomposes into coherent and incoherent components via the Rayleigh roughness factor [5] $\eta = \exp(-(2k\sigma_h \cos\theta_t)^2)$, $k = 2\pi/\lambda$, where θ_t is the incidence angle between $\boldsymbol{\omega}_t$ and the surface normal:

$$f = A(\boldsymbol{\omega}_t)\, \Big[\eta\, f_{\mathrm{coh}} + (1-\eta)\, f_{\mathrm{inc}}\Big], \tag{5}$$

where $A(\boldsymbol{\omega}_t) \in [0,1]$ is the slab Fresnel energy gate. The lobe decompositions are:

$$f_{\mathrm{coh}} = \tau\, f_{\mathrm{KA}} + (1{-}\tau)\, f_{\mathrm{SPM}}, \tag{6}$$

$$f_{\mathrm{inc}} = \gamma\, f_{\mathrm{dir}} + (1{-}\gamma)\, f_{\mathrm{broad}}, \tag{7}$$

where τ is a learnable KA/SPM blend and γ is a roughness-dependent sigmoid. The coherent lobes are: f_{KA}, a GGX microfacet specular lobe [48] with roughness $\alpha = \sqrt{4\pi\sigma_h/\lambda}$; and f_{SPM}, a von Mises–Fisher (vMF) distribution centered on the specular direction with concentration $\kappa = \sqrt{l_c/\lambda}\,/\,(1+3\sigma_h/l_c)$ [45]. The incoherent lobes are: f_{dir}, a broader vMF lobe; and f_{broad}, Lambertian. The energy gate $A(\boldsymbol{\omega}_t)$ uses an ITU-R P.2040 slab model with thickness d and complex permittivity $\varepsilon_r = \varepsilon_r' - j\varepsilon_r''$, capturing multiple internal reflections rather than single-interface Fresnel.

The Fresnel coefficients in the energy gate also determine polarization state, which we maintain throughout path tracing using a local s/p basis. At each interaction, the complex Fresnel coefficients derived from learnable permittivity update both amplitude and phase of the s and p field components [7]. Between bounces, the polarization basis is transported deterministically to preserve phase consistency. The polarization-dependent energy gate is $A = f_s|r_s|^2 + f_p|r_p|^2$, where r_s, r_p are the complex Fresnel reflection coefficients for the s and p polarizations and f_s, f_p are the antenna-dependent power fractions in each polarization.

3.3 Ray Generation

The MC estimator (Eq. (3)) requires sampling ray directions at each interaction. Because the mmWave BSDF (Sec. 3.2) combines smooth diffuse, near-delta specular, and edge-diffraction components that no single sampling PDF can efficiently cover, we provide the PDFs ρ_ℓ via three complementary strategies combined with multiple importance sampling (MIS) [47] and balance heuristic weights.

RX-centric reservoir sampling. Because mmWave antennas are highly directional, we employ RX-centric reservoir sampling inspired by Resampled Importance Sampling (RIS) [44] and spatiotemporal reservoir resampling (ReSTIR) [6, 32]. For each RX element, we sample M candidate directions from a cosine-weighted hemisphere proposal centered on the RX boresight, $\rho_{rx}(\boldsymbol{\omega}) = \cos\theta/\pi$, which naturally concentrates samples within the antenna main lobe, then retain K directions that successfully intersect scene geometry.

Specular manifold sampling. LiDAR-derived meshes contain triangles much smaller than the mmWave wavelength ($\lambda \approx 3.9$ mm). The image method (mirroring TX across the hit triangle's plane) frequently fails because the exact specular point falls outside the original triangle. We instead use Specular Manifold Sampling (SMS) [52], which solves for exact reflection points via Newton iteration on the half-vector constraint $\mathbf{C}(\mathbf{x}) = [\mathbf{s}\cdot\mathbf{h},\ \mathbf{t}\cdot\mathbf{h}]^\top = \mathbf{0}$, where $\mathbf{s}, \mathbf{t}$ are surface tangents and $\mathbf{h}$ is the half-vector. After each Newton step, re-projection onto the mesh via ray tracing allows the solver to cross triangle boundaries, producing exact specular paths with deterministic Fresnel weights.

We extend SMS for radar inverse rendering with: GPU-vectorized Newton iteration via DrJit [21], Implicit Function Theorem (IFT) gradient support that stores the Jacobian inverse at convergence for backpropagation through vertex positions, and multi-bounce specular chain solving via a block-tridiagonal algorithm.

Free-space diffraction BSDF. Unlike the Uniform Theory of Diffraction (UTD), which requires explicit edge detection and separate ray classes, the free-space diffraction (FSD) BSDF [41] formulates edge diffraction as a standard scattering distribution compatible with path tracing. At each intersection, the FSD BSDF projects nearby triangles onto a virtual screen, identifies boundary edges, and evaluates closed-form Fraunhofer diffraction integrals. A fraction β of reflected energy is redirected into importance-sampleable diffracted lobes.

We extend FSD for mmWave radar with: (i) material-dependent edge opacity via Fresnel reflectance from learnable permittivity; (ii) Jones polarization tracking through diffracted paths for MIMO phase coherence; and (iii) edge suppression filters that reject artifacts from noisy LiDAR tessellation.

3.4 Differentiable Optimization

We optimize per-vertex ITU physics materials ($6 \times N_v$ parameters with sigmoid/exponential reparameterizations for physical bounds), per-vertex normals $\mathbf{N} \in \mathbb{R}^{N_v \times 3}$, and antenna beam patterns. The pipeline additionally supports differentiable vertex positions and 6-DoF radar pose, but we keep these fixed in our experiments: our coarse-to-fine LiDAR-radar alignment (Sec. B.2 in the supplement) already provides sub-centimeter translational and sub-degree rotational accuracy, so the remaining sim-to-real gap is dominated by material and scattering properties rather than geometric misregistration. Initialization uses ITU-R P.2040 reference values for common materials (concrete, glass, metal). Details on bounds, learning rates, and schedules are in the supplement (Sec. B.4).

The entire forward model (Secs. 3.1–3.3) executes inside a single DrJit [21] computation graph with automatic differentiation. Gradients flow from $\mathcal{L}$ through the ADC, per-path phasors, BSDF evaluations, and ray–triangle intersections back to all learnable parameters in a single backward pass. At silhouette edges where visibility changes discontinuously, projective boundary gradient sampling [53] corrects gradient bias for vertex positions and pose when these are enabled.

Loss function. We minimize a range–azimuth reconstruction loss on min–max normalized magnitudes:

$$\mathcal{L}_{\text{RA}} = \frac{1}{|\Gamma|} \sum_{(r,a)\in\Gamma} \left\| \widetilde{\text{RA}}_{\text{render}}(r,a) - \widetilde{\text{RA}}_{\text{gt}}(r,a) \right\|_2^2, \tag{8}$$

where Γ is the set of range–azimuth bins, $\widetilde{\text{RA}} = (\text{RA} - \min \text{RA})/(\max \text{RA} - \min \text{RA})$ denotes min–max normalization, and RA maps are obtained via the TDM-MIMO FFT pipeline [30]. Operating directly on linear magnitudes gives uniform gradient weighting across the dynamic range, avoiding the gradient imbalance that log-compression introduces near the noise floor. The total objective

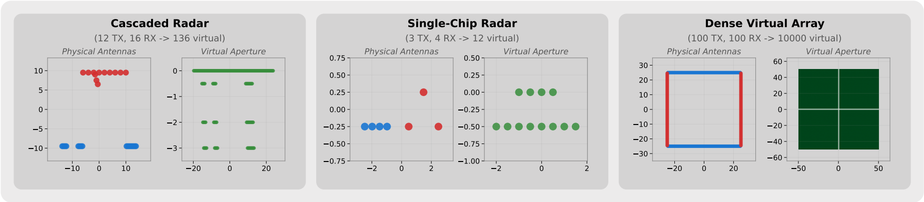

Fig. 3: Antenna configurations used in our evaluation. Left: cascaded MMWCAS-RF-EVM (12TX×16RX → 86 unique azimuth virtual elements; Sec. A.5 in the supplement) used for training. Center: single-chip AWR1843 (3TX×4RX → 12 virtual elements) used for cross-sensor transfer. Right: dense virtual array (100TX×100RX → 10,000 virtual elements) used for 3D inference. Top row: physical TX (red) and RX (blue) positions. Bottom row: virtual aperture (TX+RX convolution).

Table 2: Range–azimuth (RA) and ADC training evaluation across seven outdoor scenes. mmIR is compared against Sionna-RT [18]. Metrics: Pearson correlation (Corr), PSNR, SSIM, RMSE, and ADC-level magnitude error. Best in **bold**.

	RA Correlation ↑		RA PSNR ↑		RA SSIM ↑		RA RMSE ↓		ADC Log-Mag MSE ↓		ADC MM-Mag MSE ↓	
Scene	Ours	Sionna	Ours	Sionna	Ours	Sionna	Ours	Sionna	Ours	Sionna	Ours	Sionna
S0F135	**0.921**	0.418	**37.4**	21.4	**0.938**	0.509	**0.0135**	0.0850	**48.3271**	244.8361	**0.0421**	0.0469
S0F390	**0.936**	0.242	**41.0**	23.4	**0.937**	0.550	**0.0089**	0.0674	**5.0250**	294.3032	**0.0289**	0.0417
S1F185	**0.954**	0.312	**39.8**	18.9	**0.926**	0.326	**0.0102**	0.1138	**32.3385**	282.6453	**0.0398**	0.0461
S1F438	**0.942**	0.346	**37.7**	19.9	**0.926**	0.401	**0.0130**	0.1008	**32.5274**	288.9782	**0.0369**	0.0435
S2F105	**0.887**	0.188	**34.6**	28.7	**0.894**	0.724	**0.0185**	0.0367	**23.5161**	244.9075	0.0498	**0.0481**
S2F160	**0.864**	0.370	**34.0**	22.4	**0.789**	0.519	**0.0200**	0.0763	**27.6683**	275.3327	**0.0444**	0.0461
S2F300	**0.930**	0.388	**38.7**	21.3	**0.906**	0.512	**0.0117**	0.0858	**30.0014**	247.1371	**0.0370**	0.0543
Mean	**0.919±0.030**	0.324±0.076	**37.6±2.4**	22.3±3.0	**0.902±0.049**	0.506±0.115	**0.0137±0.0038**	0.0808±0.0229	**28.4863±11.9698**	268.3057±20.3703	**0.0399±0.0061**	0.0467±0.0037

is $\mathcal{L} = \mathcal{L}_{\text{RA}} + \lambda_{\text{geom}}\mathcal{L}_{\text{geom}}$, where λ_{geom} is a regularization weight and $\mathcal{L}_{\text{geom}}$ enforces geometric smoothness via Laplacian regularization on vertex normals and positions. After convergence, we freeze optimized parameters and re-render from dense virtual apertures to synthesize 3D radar data (Sec. 4.3).

4 Experiments

We evaluate mmIR on seven diverse outdoor scenes from the ColoRadar dataset [26], which provides synchronized mmWave radar and LiDAR captures with raw ADC access. We design three experiments that progressively test our forward model under increasingly challenging conditions: (1) training evaluation on the cascaded imaging radar used for optimization (12TX×16RX, 86 virtual elements); (2) cross-sensor transfer to a co-located single-chip radar (3TX×4RX, 12 virtual elements) without re-training; and (3) 3D occupancy inference via a dense virtual array (100TX×100RX, 10,000 virtual elements). Figure 3 illustrates all three antenna configurations. Dataset details, scene geometry reconstruction, and implementation hyperparameters are provided in the supplement (Sec. B).

4.1 Training Evaluation: ADC Rendering and Range–Azimuth

We compare rendered and measured range–azimuth (RA) maps on the cascaded radar used for optimization. Table 2 reports Pearson correlation, PSNR, SSIM,

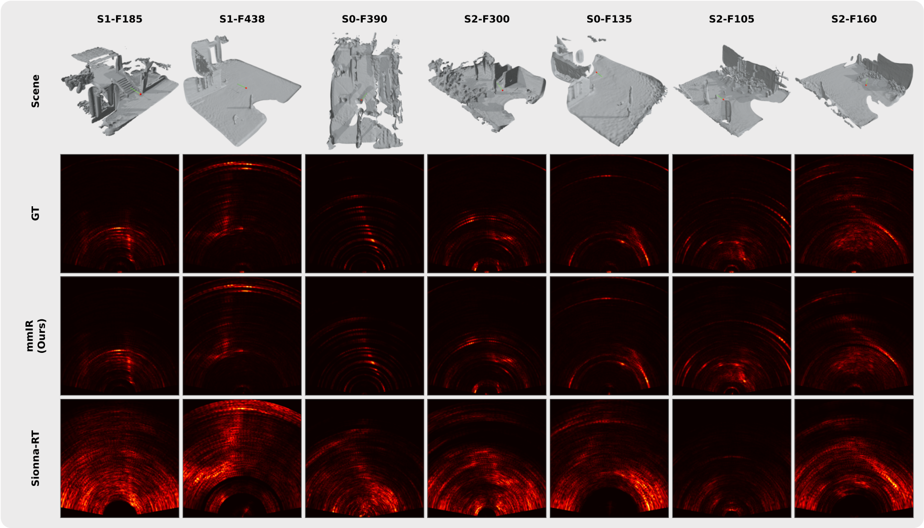

Fig. 4: Qualitative range–azimuth (RA) comparison across all seven training scenes, sorted by Pearson correlation (highest first). Each column shows one scene. Rows, top to bottom: scene mesh with radar position (red sphere) and boresight direction (green arrow); ground-truth RA; mmIR (ours); Sionna-RT baseline. Horizontal axis: cross-range (m); vertical axis: range (m). mmIR captures the dominant scatterer structure and multipath returns across all scenes, while Sionna-RT produces weaker, less spatially accurate responses.

RMSE, and ADC-level magnitude metrics across all seven scenes, with Sionna-RT [18] as baseline. Evaluating in the RA domain implicitly validates phase accuracy: the azimuth FFT converts relative phase differences across virtual elements into angular information, so any systematic phase error (per-element bias, incorrect path-length scaling, or incoherent phasor accumulation) would smear or shift targets in the angular domain, degrading all RA metrics. Absolute phase is not meaningful in practice, as it depends on user-configurable initial phase, oscillator drift, and phase noise; these common-mode offsets appear in the DC term of the spatial FFT and are dropped during processing.

mmIR achieves a mean Pearson correlation of **0.919**, substantially outperforming Sionna-RT (0.324). Table 2 quantifies this gap across all metrics; Figure 4 provides the corresponding qualitative comparison, sorted by correlation. mmIR reproduces both the dominant scatterer locations and the diffuse multipath structure present in the ground truth, whereas Sionna-RT produces attenuated responses with spatially misaligned features.

Three architectural differences drive this gap. First, mmIR's physically-based BSDF with six per-vertex learnable parameters captures spatially varying reflectance that Sionna-RT's per-object scattering model cannot represent; this is most evident in geometrically complex scenes (S1-F438, S0-F135) where mixed-material surfaces dominate the radar response. Second, mmIR jointly optimizes

Table 3: Cross-sensor transfer evaluation. Scene parameters optimized on the cascaded radar (12TX×16RX, 86 unique azimuth virtual elements) are frozen and used to render ADC for the co-located single-chip radar (3TX×4RX, 12 virtual elements) without re-training. Best in **bold**.

	RA Correlation ↑		RA PSNR ↑		RA SSIM ↑		RA RMSE ↓		ADC Log-Mag MSE ↓		ADC MM-Mag MSE ↓	
Scene	Ours	Bench.	Ours	Bench.	Ours	Bench.	Ours	Bench.	Ours	Bench.	Ours	Bench.
S0F135	**0.533**	0.002	**25.6**	17.8	**0.505**	0.464	**0.0524**	0.1283	**10.9639**	301.4341	0.2194	**0.0668**
S0F390	**0.464**	0.328	21.2	**24.2**	0.529	**0.566**	0.0873	**0.0619**	**12.9644**	331.2549	**0.0630**	0.0840
S1F185	**0.567**	0.136	**22.4**	20.1	0.422	**0.480**	**0.0759**	0.0990	**0.5754**	324.8441	0.1993	**0.0771**
S1F438	**0.684**	0.079	**23.2**	13.5	**0.674**	0.202	**0.0694**	0.2119	**21.8672**	348.0237	0.1507	**0.0662**
S2F105	**0.608**	0.084	**24.7**	22.8	0.609	**0.690**	**0.0583**	0.0721	**3.4458**	289.7929	0.1959	**0.1068**
S2F160	**0.276**	0.159	**19.3**	18.3	**0.493**	0.463	**0.1085**	0.1219	**20.1530**	338.1699	0.0893	**0.0798**
S2F300	**0.745**	0.008	**27.2**	23.6	**0.468**	0.412	**0.0434**	0.0661	**7.7226**	287.9379	0.1074	**0.0948**
Mean	**0.554±0.143**	0.114±0.103	**23.4±2.5**	20.0±3.6	**0.529±0.080**	0.468±0.138	**0.0707±0.0206**	0.1087±0.0488	**11.0989±7.3886**	317.3511±22.3659	0.1464±0.0565	**0.0822±0.0136**

surface normals and antenna beam patterns, parameters entirely outside Sionna-RT's optimization scope, correcting both surface orientation errors inherited from noisy LiDAR tessellation and systematic antenna gain biases. Third, Sionna-RT's path solver executes outside the AD graph: ray tracing runs under gradient suspension and all path geometry is subsequently detached, so while material gradients propagate through field coefficients at each bounce, the path topology itself is frozen and no gradients reach vertex positions or normals, and a material change at one surface cannot redirect subsequent bounces. mmIR places the entire forward model inside a single computation graph, enabling end-to-end gradient flow from the loss through ADC synthesis, phasor accumulation, BSDF evaluation, and ray–triangle intersections. The weakest scene, S2F160 (0.864), shows that even challenging geometries with complex multipath achieve high correlation. mmIR substantially outperforms the baseline on every scene, indicating that the richer parameter space and deeper gradient flow consistently improve fit quality.

4.2 Radar Transfer Evaluation

A central claim of physics-based inverse rendering is that the optimized scene captures true physical properties, not sensor-specific patterns. We validate this by *cross-sensor transfer*: the scene parameters trained on the cascaded radar (12TX × 16RX, 86 virtual elements) are frozen and used to render ADC for the single-chip radar (3TX × 4RX, 12 virtual elements) that shares the same physical mounting (Fig. 3). We apply the same rigid-body alignment transformation learned during cascaded training to the single-chip configuration and render forward-only.

This experiment tests whether our inverse renderer has learned transferable scene physics rather than memorizing the training sensor's idiosyncrasies. Table 3 shows that mmIR achieves a mean transfer correlation of **0.554**, outperforming Sionna-RT (0.114), demonstrating that the learned materials generalize across sensor configurations.

Table 4: 3D occupancy evaluation against LiDAR ground truth. Baseline: ColoRadar-provided 3D point cloud from the cascaded radar's near-1D virtual aperture, where elevation is synthesized by extruding range–azimuth detections along the vertical axis due to minimal elevation resolution. Ours: dense virtual array (100TX×100RX) with full 2D aperture enabling range–azimuth–elevation (RAE) imaging. Occupancy metrics use KD-tree nearest neighbors (τ=0.5 m). Best in **bold**.

	Precision ↑		Recall ↑		F1 ↑		Accuracy ↑		RMSE ↓		R-CD ↓	
Scene	Ours	ColoR	Ours	ColoR	Ours	ColoR	Ours	ColoR	Ours	ColoR	Ours	ColoR
S0F135	0.077	**0.223**	0.018	**0.054**	0.029	**0.087**	0.044	**0.057**	**2.324**	3.302	0.248	**0.228**
S0F390	**0.481**	0.045	**0.110**	0.014	**0.179**	0.021	**0.212**	0.017	**2.692**	2.745	0.216	**0.207**
S1F185	**0.997**	0.483	**0.566**	0.082	**0.722**	0.141	**0.703**	0.103	**0.909**	2.380	**0.067**	0.236
S1F438	**0.633**	0.433	**0.265**	0.117	**0.373**	0.185	**0.362**	0.131	**1.511**	3.339	**0.116**	0.268
S2F105	**0.771**	0.154	**0.162**	0.025	**0.268**	0.044	**0.340**	0.029	**1.099**	3.402	**0.113**	0.282
S2F160	**0.516**	0.383	**0.177**	0.165	**0.264**	0.231	**0.257**	0.183	1.917	**1.611**	0.146	**0.118**
S2F300	**0.850**	0.508	**0.501**	0.122	**0.630**	0.197	**0.601**	0.130	**1.040**	1.643	**0.083**	0.118
Mean	**0.618±0.278**	0.319±0.165	**0.257±0.189**	0.083±0.051	**0.352±0.228**	0.129±0.075	**0.360±0.210**	0.093±0.056	**1.642±0.638**	2.632±0.721	**0.141±0.062**	0.208±0.061

4.3 Inference: 3D Occupancy via Dense Virtual Array

After training, we freeze the optimized scene parameters and re-render from a dense virtual array (100TX × 100RX, 10,000 virtual elements with $\lambda/2$ spacing) inspired by the Rohde & Schwarz QAR50 [51] (Fig. 3, right). We convert synthesized ADC into a 3D tensor via Range–Azimuth–Elevation (RAE) FFT and extract a sparse radar point cloud by thresholding at the 99.9th intensity percentile. Following [27], we use single-bounce propagation at test time for the dense array.

We evaluate against LiDAR after radar-guided filtering: (1) filtering by the dense array's 0 dB beam limits, (2) projecting LiDAR points onto the radar polar voxel grid, and (3) retaining only points where radar normalized intensity ≥ 0.1. Occupancy metrics use KD-tree nearest neighbors with $\tau = 0.5$ m threshold; geometric fidelity is measured via RMSE and Relative Chamfer Distance (R-CD). Table 4 shows substantial variation driven by radar visibility: scenes with multiple surfaces within the field-of-view achieve strong reconstruction (S1F185: Precision=0.997, R-CD=0.067), while open scenes with ground planes nearly parallel to boresight show degraded recall as specular energy is directed away from receivers. Additional qualitative analysis, quantitative ablation, and runtime results are in the supplement (Sec. C).

5 Conclusion

We introduced mmIR, a differentiable inverse-rendering framework that fits scene and sensor parameters directly from raw FMCW ADC. By combining end-to-end automatic differentiation through phase-coherent MIMO path accumulation with ITU physics-based per-vertex materials, mmIR optimizes materials, normals, and beam patterns from real radar captures. On seven outdoor scenes, mmIR achieves 0.63 mean Pearson correlation on range–azimuth maps, substantially outperforming Sionna-RT (0.32). After fitting, the optimized scene produces sensor-realistic ADC from dense virtual apertures for 3D occupancy

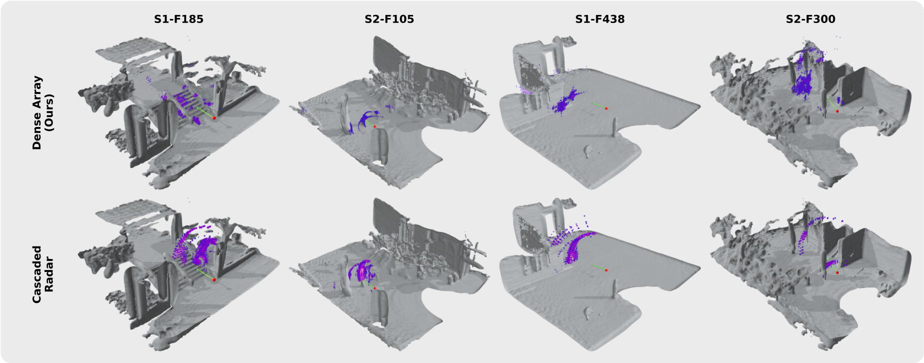

Fig. 5: Single-frame 3D occupancy comparison overlaid on the LiDAR mesh. Top row: mmIR dense virtual array (100×100 elements, uniform $\lambda/2$ spacing) with full 2D aperture enabling range–azimuth–elevation (RAE) imaging, thresholded at the 99.9th intensity percentile. Bottom row: ColoRadar-provided 3D point cloud from the cascaded radar's near-1D virtual aperture, which resolves only range–azimuth and extrudes detections along elevation, producing vertically smeared structure. Red sphere: radar position; green arrow: boresight direction.

detection and scene reconstruction, validated against LiDAR. We further validate cross-sensor generalization by transferring trained scenes to a different radar configuration. This opens a path toward physics-grounded radar sensing pipelines, including calibration, radar-aware 3D reconstruction, and joint multi-modal optimization.

Limitations and future work. mmIR is bottlenecked by mesh quality: the LiDAR-to-mesh reconstruction pipeline sets a ceiling on rendered ADC realism, and the inverse rendering task is sensitive to LiDAR-radar alignment for both cascade and single-chip configurations. Our prototype further assumes static scenes and surface-only interactions (no transmission, refraction, or volumetric scattering), models diffraction at single-bounce interactions only, and does not account for Doppler phase from ego-motion. Although our pipeline supports differentiable vertex positions and pose, we found that at 77 GHz ($\lambda \approx 3.9$ mm), sub-millimeter vertex shifts induce large phase changes, making geometry and pose gradients unstable under Monte Carlo estimation with current sample budgets. Shape-from-radar, where geometry is directly recovered from radar measurements, remains a compelling goal but will likely require dense multi-view supervision and regularization strategies beyond what single-frame RA fitting provides. Natural extensions include higher-fidelity meshing, robust alignment, temporally coherent Doppler modeling, multi-bounce diffraction, and richer propagation effects. We discuss these limitations in detail in the supplement (Sec. D). Scaling to jointly optimize radar with other sensors and learning priors over radar-consistent materials are promising directions for robust autonomous perception.

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